EDA summary report

1. Introduction:

This document presents an exploratory analysis of Geldium's dataset, aimed at evaluating data integrity, uncovering valuable insights, and identifying factors that contribute to the risk of credit default. The primary objective is to prepare the data for accurate predictive modeling and risk evaluation.

2. Dataset overview:

The dataset comprises 500 customer records with 19 variables and is intended for the prediction of account delinquency. It integrates demographic information (age, location, employment status), financial attributes (income, loan balance, debt-to-income ratio, and credit utilization), credit profile indicators (credit score, credit card type, and account tenure), and behavioural measures of repayment performance (number of missed payments and monthly payment status over a six-month period).

Key dataset attributes:

- Number of records – 500 customers
- no. of **Columns:** 19 features
- **Target variable:** `Delinquent_Account` (binary)
- Data types:
  - Categorical: Employment Status, Credit Card Type
  - Numerical: Income, Loan Balance

3. Missing Data Analysis:

The pattern of missing data is not uniform across variables and is likely non-random, as incomplete income and loan information may be associated with customers having unstable employment, limited credit history, or reporting constraints.

Missing values

column	Missing count	%
Income	39	~7.8%
Loan Balance	29	~5.8%
Credit Score	2	~0.4%
Others	0	0%

Median imputation was applied to *Income*, *Loan Balance*, and *Credit Score* to address missing values, as these variables exhibit skewed distributions and the median provides a robust estimate suitable for logistic regression.

- Fields with missing data:
  - Income: 50 missing entries
  - Loan Balance: 30 missing entries
- Planned solutions:
  - Use the median to fill missing numeric values
  - Apply AI-generated synthetic data where appropriate for Loan Balance

4. Key Findings and Risk Indicators

The analysis identifies several variables with strong associations to account delinquency, reflecting both structural credit risk and short-term behavioural patterns.

- **Credit Score:** Lower credit scores are strongly associated with higher delinquency risk, making this one of the most influential predictors in the dataset.

- **Credit Utilization:** Higher utilization ratios indicate financial stress and are linked to an increased likelihood of delinquent accounts.

- **Debt-to-Income Ratio:** Elevated ratios reflect reduced repayment capacity and contribute to higher delinquency risk.

- **Account Tenure:** Customers with shorter account tenures exhibit a greater probability of delinquency, suggesting limited credit history and stability.

- **Missed Payments:** An increased number of missed payments is one of the strongest behavioural indicators of delinquency.

- **Recent Payment Behaviour:** Late or missed payments in recent months significantly raise delinquency risk, while consistent on-time payments are associated with non-delinquent accounts.

Important insights:

- Customers using more than 50% of their credit limit tend to be at greater risk.
- Individuals with 3 or more missed payments within six months show a higher likelihood of defaulting.
- Some inconsistencies were observed: high-income customers with low credit scores warrant further examination.

5. Role of AI & GenAI

Generative AI tools supported the identification of trends, detection of missing values, and examination of risk elements. These AI-based conclusions were compared against established financial risk metrics for validation.

Sample AI queries:

- "Summarize data trends and highlight missing values."
- "Assess risk of default based on credit usage and payment behavior."

6. Conclusion & Future Actions

This EDA uncovered meaningful insights into Geldium's dataset, highlighting missing entries, behavioral patterns tied to credit risk, and some outlier cases worth deeper analysis.

Takeaways:

- Data gaps: Missing income and loan data could influence outcomes.
- Delinquency indicators: High credit usage and repeated missed payments are strong predictors.
- Data anomalies: Cases of high income but low credit scores need clarification.

Recommendations:

- Choose suitable imputation techniques for missing income and loan values to minimize bias.
- Confirm if key risk factors remain consistent across various customer groups.

- Investigate irregular data entries to ensure accuracy and detect potential financial instability.

These efforts will aid Geldium in refining its risk analysis processes and enhance data reliability for further modeling.